



Reliability analysis of compressive strength of brick, conventional concrete and geopolymer concrete by using ANN

Manvendra Verma¹

Received: 11 August 2025 / Accepted: 19 August 2025 / Published online: 28 August 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract

This study investigates the reliability estimation of the compressive strength of brick, conventional concrete, and geopolymer concrete cubes through the application of non-destructive tests (NDTs). Non-destructive testing methods, such as rebound hammer and ultrasonic pulse velocity (UPV) tests, offer a rapid and practical approach for assessing the structural integrity and strength of materials without causing damage. The research evaluates the effectiveness of these methods in estimating the compressive strength of the three different materials brick, conventional concrete, and geopolymer concrete and compares the results with conventional destructive testing (CDT) methods like the compressive strength test on cube specimens. Statistical analysis is employed to establish correlations between NDT results and actual compressive strength values, providing insight into the reliability and accuracy of NDT techniques for different materials. The findings demonstrate that NDTs can serve as reliable alternatives to CDTs, though the level of accuracy varies depending on the material type. The study concludes that while NDT methods are highly effective for preliminary evaluations and in-situ assessments, further calibration may be required for different materials to ensure precise compressive strength estimation.

Keywords Non-destructive tests · Geopolymer concrete · Bricks · ANN · RBF classifier

Introduction

The properties of the concrete being used in structures are what need to be determined by doing non-destructive in-place testing on the concrete. The ability to compress something is commonly sought after (Guo et al., 2010). In order to provide an accurate estimate of strength, there has to be a well-established link between the outcomes of in-place testing and the strength of concrete. In most cases, an attempt to assess this connection is made in the laboratory (Verma, 2021). The degree of association that exists between the strength of the concrete and the amount of in-place testing that is carried out is directly proportional to the accuracy of the strength prediction (ACI 228.1R-03, 2003). Two of the most used non-destructive testing (NDT) techniques for determining the strength of concrete are the rebound and ultrasonic pulse velocity (UPV) measurements. A new

guideline provides a formal solution for predicting concrete strength from in situ testing (Abd et al., 2014).

A crucial issue that has to be addressed is the formulation and verification of a technique that, when used, yields a trustworthy assessment of strength together with an appropriate degree of assurance. “Calibration,” which refers to the process of building and using a reliable relationship between NDT results and strength, is an essential component. Calibration is an essential ingredient (Qaswari, 2000). According to what is mentioned in the ACI MONOGRAPH Series, the rebound readings will be lower if the concrete specimens are on the smaller side. Any movement that is generated by the impact will have this effect. In these kinds of situations, the specimen has to be held in place or propped up by a significant amount of weight. It is advised that the testing equipment be used to get a firm grasp on the specimen. Mitchell and Hoagland revealed that the retention load seems to be around 15% of the specimen’s maximum strength at the point at which the rebound number stays constant (Kheder, 1999).

In the current investigation, a quarter of the rock specimens’ maximum strength was taken into consideration. When used jointly, UPV and rebound provide results that

✉ Manvendra Verma
Manvendra.verma@gla.ac.in

¹ Department of Civil Engineering, GLA University, Mathura, India

are more trustworthy and that are closer to the real values than when each method is used alone to make a forecast of the strength of concrete (Breysse, 2012). However, neither method is suitable for making an accurate prediction of the strength of concrete when used alone. The findings are inconsistent as a direct consequence of the fact that the combined strategy only yielded moderate benefits. There have been many different connections proposed in order to estimate the power based on a small number of data (UPV, rebound). It does not seem that there is a distinct link, and the question of calibration continues to be a crucial one, just as it does with each particular approach (Breysse, 2012).

Limestone was the predominant construction material for large building projects until reinforced concrete buildings became the norm in the construction industry. The vast majority of old and antique constructions were built using stones and bricks as the primary building materials. For instance, the fundamental structural element of these buildings in ancient Egypt was a series of columns with bases formed of a specific kind of rock such as marble, basalt, granite, or limestone (Szilágyi et al., 2011; Sreenivasulu et al., 2020). This was one of the key architectural features of these structures. The use of core samples to evaluate the compressive strength of certain rock types is prohibited by the governments of several countries (Turgut et al., 2008). When these structures are being repaired or inhabited, this operation is necessary and cannot be skipped. As a direct consequence of this, non-destructive testing is the one and only technique that is authorized for determining the compressive strength of certain materials (Rodríguez de Sensale, Rodríguez Viacava and Aguado, 2016; Mohan & Poobal, 2018).

The number of test specimens that are needed for compressive strength evaluation may be cut down significantly with the help of certain novel architectures as well as non-destructive compressive strength estimate. In the case of refractory bricks, for instance, the ASTM C 133 standard advises testing ten bricks for every one thousand bricks in order to guarantee the compressive strength of this kind of brick. This quantity of bricks may not be adequate in certain circumstances owing to the importance of the construction or the risk it poses (Kheder, 1999; Mukharjee & Barai, 2020). One example of these kinds of constructions is the chimneys found in power plants, which place a significant amount of importance on the superiority of the bricks used in their construction. Therefore, the number of specimens for compressive strength tests needs to be increased in this scenario. Alternatively, in accordance with ASTM C 133, the same number of specimens can be used to establish a relationship between compressive strength and other non-destructive in-place tests in order to estimate the compressive strength for an additional number of bricks without

Table 1 Ultrasonic pulse velocity, rebound hammer strength, and compressive strength of bricks

S. No.	Ultrasonic pulse velocity (km/sec)	Rebound strength (MPa)	Compressive strength (MPa)
1.	2.92	18.4	16.4
2.	2.94	18.6	17.2
3.	2.93	19.5	17.8
4.	2.96	18.2	16.2
5.	3.12	20.1	18.1
6.	2.95	20.6	18.4
7.	2.92	19.8	17.6
8.	2.94	19.6	17.5
9.	3.02	19.3	17.5
10.	3.08	20.6	18.4
11.	3.	19.1	16.9
12.	2.96	18.6	16.7
13.	2.92	18.9	16.2
14.	2.94	19.3	16.8
15.	2.93	19.8	17.9
16.	2.96	19.7	17.2
17.	3.12	20.3	17.8
18.	2.95	20.6	18.5
19.	2.92	21.2	18.9
20.	2.94	20.2	17.6
21.	3.02	19.5	16.8
22.	3.08	19.9	17.6
23.	3.	19.8	17.3
24.	2.96	19.1	16.9
25.	2.82	18.8	16.2
26.	3.21	20.6	18.9
27.	3.05	20.9	18.4
28.	3	18.2	15.9
29.	3.09	18.8	16.8
30.	3.12	20.9	18.3

actually performing compressive strength tests, and these specimens can be reused within the structure. It does not seem that there is a specific link, and the question of calibration continues to be an important one, just as it is with individual techniques (Xu et al., 2014).

Most people are familiar with the ultrasonic pulse velocity and surface hardness processes when it comes to doing non-destructive in-place inspections on concrete structures (Solís-Carcano & Moreno, 2008; Kim et al., 2009; Szilágyi et al., 2011). The time needed for the pulse of longitudinal ultrasonic waves to go through a material may be calculated using the ultrasonic pulse velocity technique. The travel periods that are measured electronically between the first commencement of the pulse and its reception are called the pulse arrival time. The average speed of wave propagation may be determined by dividing the distance between transducers by the amount of time that has elapsed from the beginning of the wave (Solís-Carcano & Moreno, 2008;

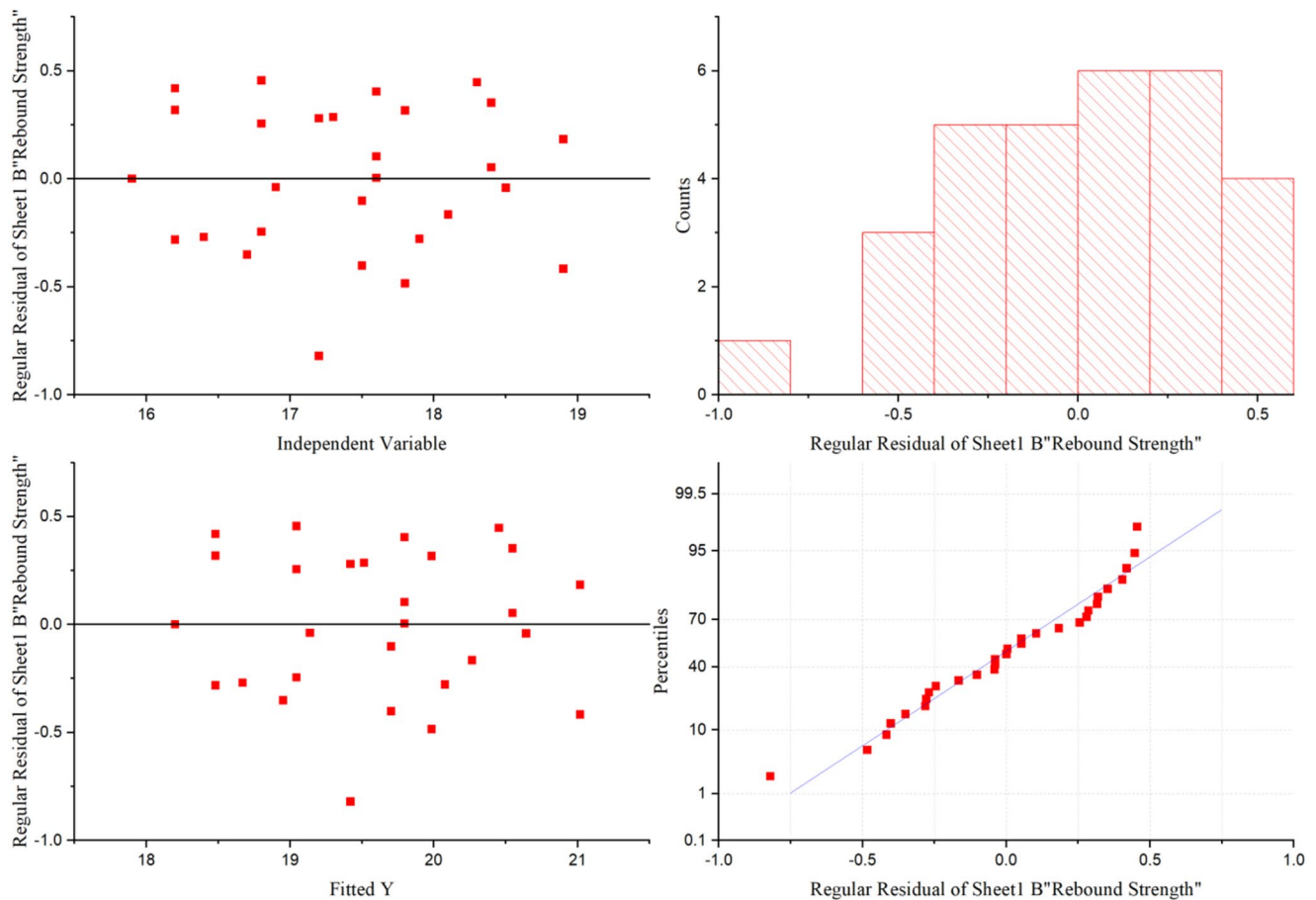


Fig. 1 Variation of rebound strength data

Tennakoon et al., 2017). The ASTM C597 standard details the appropriate equipment as well as the standard operating procedures. The ultrasonic pulse velocity test has been hailed as a useful and reliable non-destructive approach for assessing the mechanical parameters of concrete in existing

concrete structures by a number of publications. These publications have used the test as an example (Vasconcelos et al., 2008).

To assess the surface hardness of concrete, one approach includes hitting the surface with a certain quantity of energy and then determining the depth of the depression or the rebound number (Erdem et al., 2018; Varghese et al., 2019). The procedures for carrying out this test have been defined, and the relevant information may be found in ASTM standard C 805. Despite the fact that the Schmidt hammer was first developed for use with concrete, it has been put to significant use in order to provide a preliminary assessment of the stone's strength (O, Z and J-C, 2000). Artificial Neural Networks (ANNs) have several disadvantages, including being computationally expensive, requiring large amounts of data, being prone to overfitting, and lacking transparency in their decision-making processes. They also require significant expertise to design and tune, and their performance can be sensitive to input data quality.

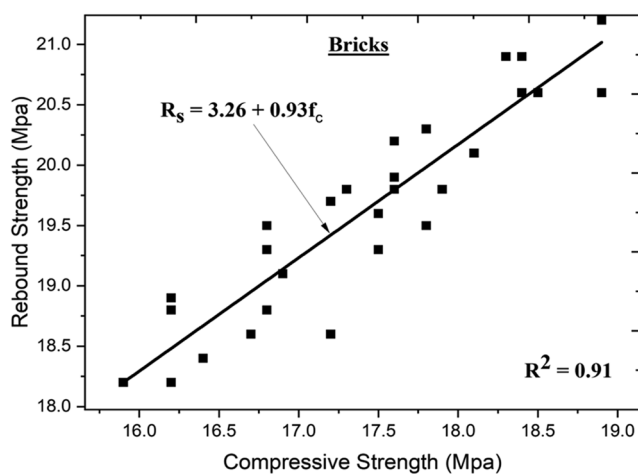


Fig. 2 Linear regression between compressive strength and rebound strength of bricks

Research significance

As was said before, trustworthy correlations were discovered to exist between brick, concrete compressive strength, and non-destructive in-place testing methods such as ultrasonic pulse velocity and surface hardness. These connections were used rather often in the process of determining the compressive strength of concrete buildings that were already in existence. Compressive strength of individual components of historic buildings or new structures composed of non-concrete materials (materials other than concrete) must be assessed in certain cases. These structures may be new or historic. When it comes to the link between non-destructive in-place testing and the compressive strength of certain construction materials, there is a dearth of data. The goal of this study is to determine whether or not there is a reliable link between the ultrasonic pulse velocity and the surface hardness (rebound number) of brick, as well as the cube compressive strength of a number of different types of construction materials. In the course of this research project, a variety of well-known building materials, such as brick, traditional concrete, and geopolymer concrete cubes, are investigated.

Materials and methods

In order to conduct non-destructive testing, bricks, concrete cubes, and geopolymer concrete cubes were used as the test materials. Both types of laboratories were responsible for casting the sample cubes. Both of these concrete cubes were intended for use with M30. After three days, seven days, 14 days, 28 days, 42 days, and 56 days, the cubes were put through their paces in a series of tests. The destructive compressive strength test is performed on the same day as the non-destructive testing, together with the destructive tests. In the non-destructive portion of the test, thirty samples of bricks were used, but only three samples of concrete cubes were utilized during the conduction portion of the test. Both the rebound hammer test and the UPVT test were carried out in accordance with the ASTM standards (ASTM C805/C805M 18, 2002; ASTM C597 16, 2016).

Procedure to tests

Step 1: - Collection of bricks, conventional concrete and geopolymer concrete samples.

Step 2: - The ASTM C170 dimension restrictions for compressive strength testing require that specimens be prepared by sawing such that they conform to cubes with a minimum dimension of 50.8 mm. The cubes were allowed

to dry out in their natural environment until the time of the tests.

Step 3: - The rebound strength and UPV was measured for each specimen in accordance with ASTM standards code.

Step 4: - The centre of the compression testing machine was loaded to roughly 25.0% of each kind of building material's ultimate compressive strength, and then specimens of each building material were introduced into the machine. The load was kept at a constant level for a certain amount of time, and then, after that, the rebound number of these specimens was calculated. Ten measurements were gathered in order to provide an estimate of the average rebound.

Step 5: - After reading the rebound number, the level of force that was being applied was increased until failure was attained. Following this step, the cube compressive strength of each specimen was determined.

Step 6: Establish a connection between the compressive strength of a material and either its rebound strength or ultrasonic pulse velocity.

Results and discussion

This section describes the results detailed that are found through the experimental tests. The non-destructive tests (rebound hammer test and ultrasonic pulse velocity test) and compressive strength test were conducted on the all samples.

Table 1 gives information on the Ultrasonic Pulse Velocity (UPV), Rebound Strength, and Compressive Strength of a material, probably concrete, at different measurement places or circumstances. These characteristics are critical for determining the quality and longevity of concrete or comparable building materials. Let us dissect the data and evaluate its significance: The first column displays each data point's serial number or identifier, basically numbering each observation for easy reference. Ultrasonic Pulse Velocity (kilometres per second): This column indicates the ultrasonic pulse velocity, which is a measure of how fast sound waves pass through a material, commonly represented in kilometres per second (km/sec). The ultrasonic pulse velocity test is a non-destructive method for determining the uniformity and integrity of concrete or other materials. Higher numbers often indicate higher material quality and uniformity. Rebound strength is a measure of the material's surface hardness or strength expressed in megapascals (MPa). A rebound hammer is used to determine this. A higher rebound strength suggests that the material is tougher and perhaps more durable. Maximum Compressive Strength (MPa): Concrete and other building materials' compressive strength is an important attribute that represents their

capacity to sustain axial loads or stresses. It's also measured in megapascals (MPa). For structural materials, high compressive strength is desirable. Let us now examine the data and make some observations: The information appears to have been gathered from 30 distinct measurements or tests. A fluctuation in the speed of sound through the material may be seen in the Ultrasonic Pulse Velocity (UPV) measurements, which range from 2.82 km/sec to 3.21 km/sec. Rebound strength varies between 18.2 MPa and 21.2 MPa, indicating that surface hardness may vary across measurements. The material can bear compressive pressures up to a maximum of 18.9 MPa, with slight variations in its compressive strength between 15.9 MPa and MPa. You may wish to use statistical analysis and visualisation methods like: to draw relevant conclusions or judgements based on this data. Statistically descriptive data to comprehend the central tendency and variability in the data, calculate the mean, median, and standard deviation for each parameter. Analysing correlations using scatter plots: Look for any trends in the correlations between UPV, rebound strength, and compressive strength. Test your hypotheses to see whether there are statistically significant differences between sets of data or if specific variables have an impact on the qualities of the material. Visualisation: Construct diagrams or charts to graphically portray the facts, making them simpler to

understand and convey. Engineers, building professionals, and material scientists can ultimately utilise the data to evaluate the construction material's quality and determine if it is appropriate for a variety of uses. Figure 1 describes the data rebound strength on bricks (Verma & Dev, 2022; Kumar et al., 2024). Figure 2 shows the regression analysis graph between compression strength and rebound strength. Figure 3 shows the variation of ultrasonic pulse velocity data. Figure 4 describes the linear regression between compressive strength and UPV of bricks.

Data for numerous metrics measured at various points or situations are included in the Tables 2 and 3. Here is a description of the information in the table. Each data point in a column is represented by a serial number or other unique identifier, simply numbering each observation for convenience. (km/sec) Ultrasonic Pulse Velocity: The Ultrasonic Pulse Velocity (UPV), measured in this column, is given in kilometres per second (km/sec). UPV is a non-destructive testing technique used to gauge how quickly sound waves go through a substance. To assess the quality and integrity of materials like concrete, it is frequently used in construction and material testing. more UPV levels often suggest more uniformity and material quality. Measurements of the Rebound Strength, expressed in megapascals (MPa), are provided in this column (Nigam & Verma, 2024). Rebound

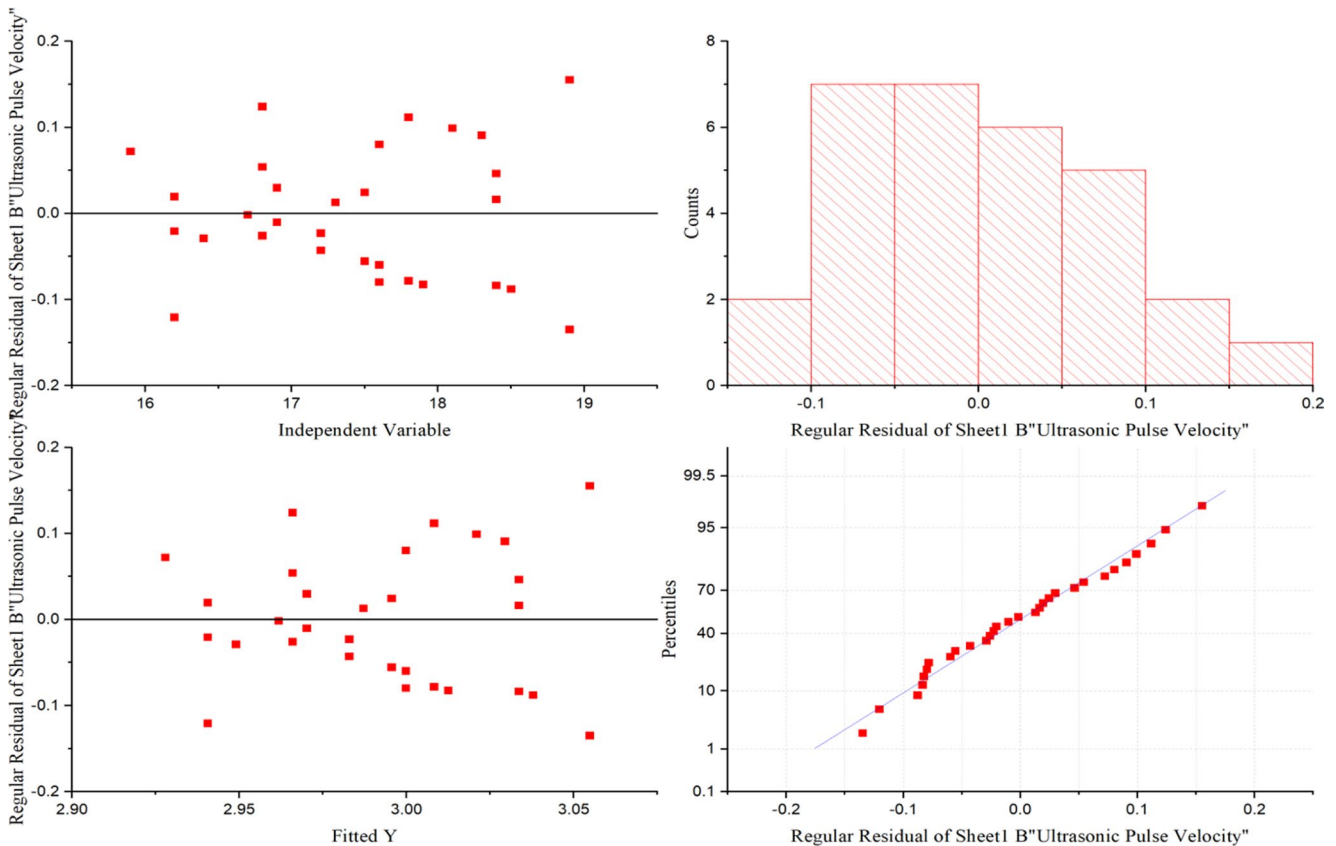


Fig. 3 Variation ultrasonic pulse velocity data

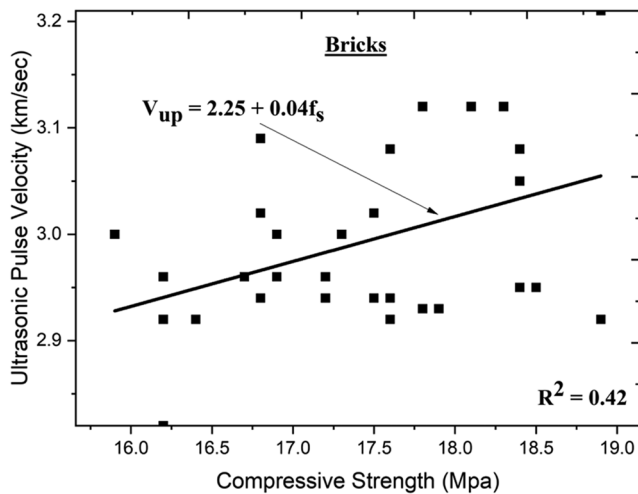


Fig. 4 Linear regression between compressive strength and UPV of bricks

Table 2 Ultrasonic pulse velocity, rebound hammer strength, and compressive strength of oven-cured geopolymer concrete

S. No.	Ultrasonic pulse velocity (km/sec)	Rebound strength (MPa)	Compressive strength (MPa)
1.	3.626	32.4	29.0
2.	4.226	35.0	32.9
3.	3.858	33.8	30.2
4.	3.764	32.7	29.3
5.	2.45	27.8	20.8
6.	3.62	30.9	27.7
7.	4.15	33.2	31.6
8.	4.23	34.4	32.6
9.	4.17	33.8	32.4
10.	4.12	32.4	31.1
11.	3.64	31.6	27.6
12.	4.27	35.4	33.2
13.	4.15	34.5	30.9
14.	3.58	32.5	28.2
15.	2.62	26.8	23.2
16.	2.98	29.8	25.0
17.	3.62	32.3	27.9
18.	4.52	35.9	34.2
19.	4.17	35.4	31.0
20.	2.86	28.1	24.6
21.	4.42	35.8	34.7
22.	4.51	35.9	35.1
23.	4.61	36.2	35.7
24.	4.29	35.5	32.5
25.	4.05	33.5	30.6
26.	4.59	40.5	38.7
27.	4.53	42.4	40
28.	4.51	43.5	38.3
29.	4.59	44.8	41.2
30.	4.49	43.3	40.1

Table 3 Ultrasonic pulse velocity, rebound hammer strength, and compressive strength of ambient-cured geopolymer concrete

S. No.	Ultrasonic pulse velocity (km/sec)	Rebound strength (MPa)	Compressive strength (MPa)
1.	3.389	20.8	18.2
2.	3.624	28.1	24.3
3.	3.503	25.7	21.8
4.	3.412	24.4	20.3
5.	2.18	21.1	13.9
6.	2.45	23.8	20.4
7.	2.82	24.5	22.8
8.	2.82	27.3	24.0
9.	2.75	26.8	23.8
10.	2.46	25.0	23.1
11.	2.38	21.4	23.2
12.	2.41	28.5	24.4
13.	2.32	25.4	23.5
14.	2.14	22.0	20.3
15.	1.52	19.4	13.9
16.	1.63	21.6	15.5
17.	2.18	23.5	21.8
18.	2.12	25.7	24.0
19.	2.93	25.9	25.0
20.	1.94	18.0	20.3
21.	2.83	25.8	25.0
22.	2.78	26.3	25.4
23.	3.12	26.9	25.8
24.	2.49	25.7	23.4
25.	2.32	25	23.5
26.	2.14	22.0	20.3
27.	1.52	19	13.9
28.	1.63	21	15.5
29.	2.18	23	21.8
30.	2.12	25	24.0

strength is a measurement of the material's surface hardness or strength that is normally obtained with a rebound hammer. A surface with higher values is likely to be tougher and more durable. Measurements of compressive strength, commonly known as megapascals (MPa), are provided in this column. The capacity of building materials to sustain axial or compressive forces without breaking is referred to as their compressive strength. Greater load-bearing capability is indicated by higher compressive strength ratings. Figure 5 describes the data rebound strength of oven-cured geopolymer concrete. Figure 6 shows the regression analysis graph between compression strength and rebound strength. Figure 7 shows the variation of ultrasonic pulse velocity data (Tiwari & Verma, 2024; Nigam & Verma, 2025). Figure 8 describes the linear regression between compressive strength and UPV of oven-cured geopolymer concrete. There are 30 rows in the table, each of which represents a distinct measurement or test situation. The information exposes changes

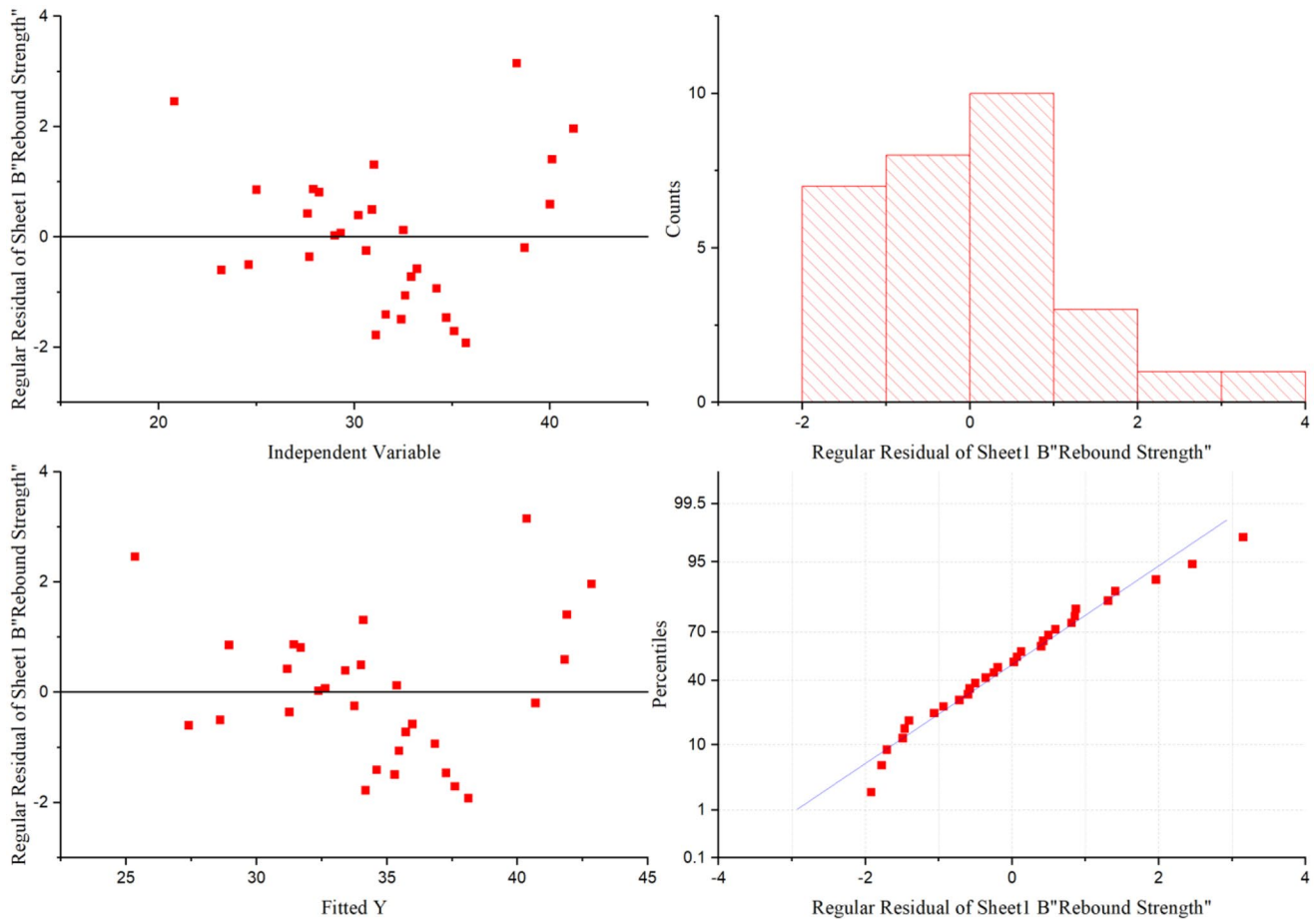


Fig. 5 Variation of rebound strength data of oven cured geopolymer concrete

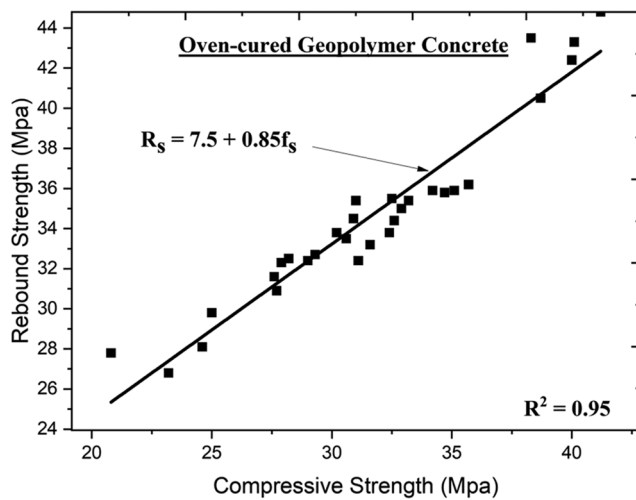


Fig. 6 Linear regression between compressive strength and rebound strength of oven-cured GPC

in a material's UPV, Rebound Strength, and Compressive Strength at various places or under various circumstances. This information can be used for quality control, material selection, or structural design by engineers, contractors, or

materials scientists. Making educated judgements regarding the material's usage in building projects may be aided by analysing trends and patterns in the data to determine if the material is suitable for a certain application. Figure 9 describes the data rebound strength of ambient-cured geopolymer concrete. Figure 10 shows the regression analysis graph between compression strength and rebound strength. Figure 11 shows the variation of ultrasonic pulse velocity data. Figure 12 describes the linear regression between compressive strength and UPV of ambient-cured geopolymer concrete.

Mathematical formula

Accuracy helps to find how many test cases are predicted correctly out of all test cases in datasets. Classifier accuracy of a test set is a percentage of test set that is correctly classified by classifier. Accuracy is measured by identifying correctness of classified results. It is computed using the equation:

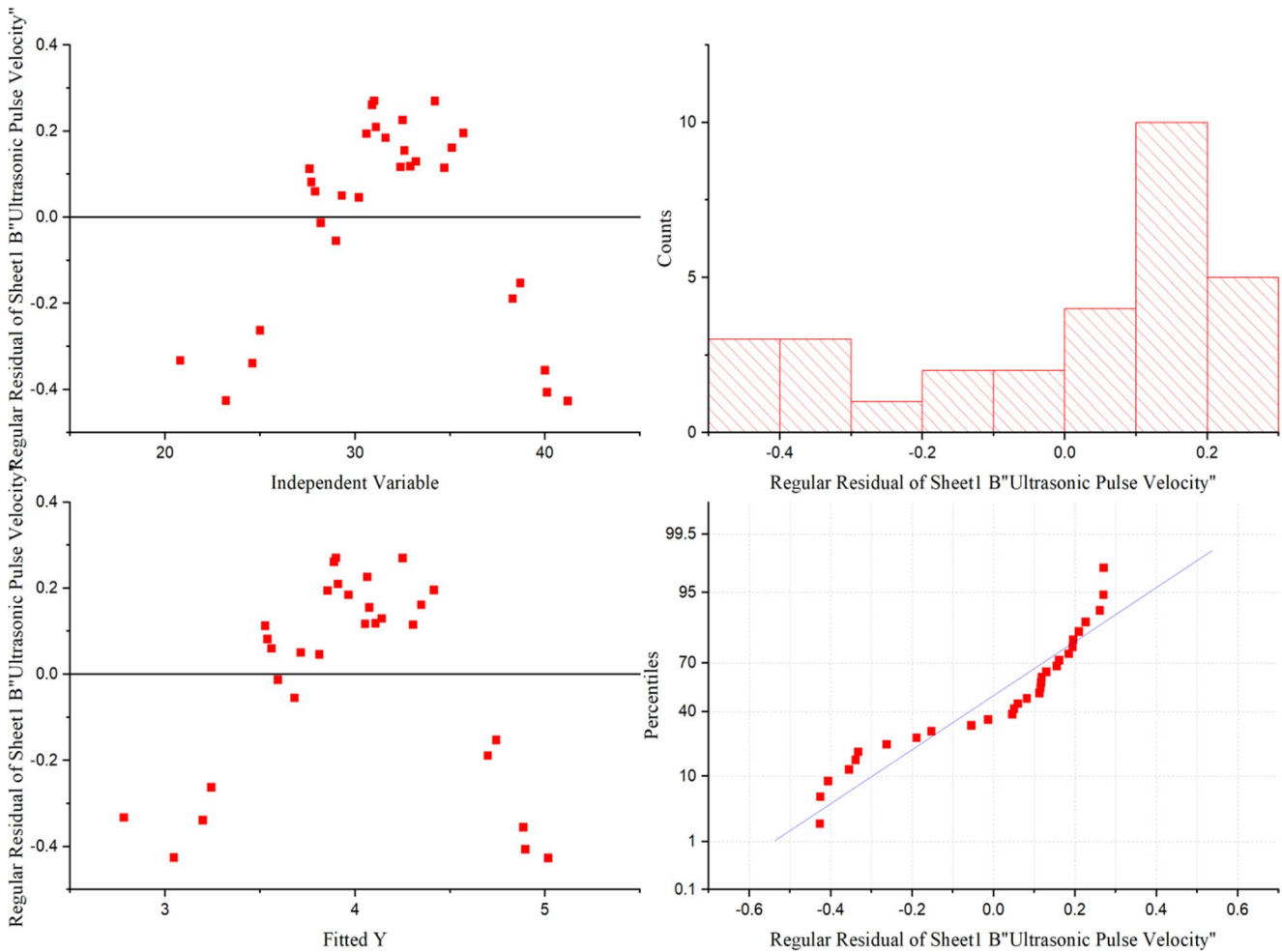


Fig. 7 Variation of ultrasonic pulse velocity of oven cured geopolymer concrete

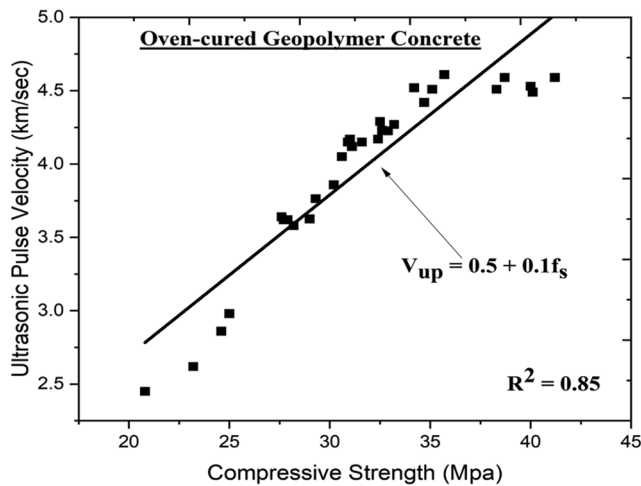


Fig. 8 Linear regression between compressive strength and UPV of oven-cured GPC

$$Accuracy = \frac{Tp + Tn}{Tp + Fn + Tn + Fp} * 100\% \quad (1)$$

Precision metric helps us to measure how many cases are estimated correctly of a certain category. It measures the correctness of the classifier. It is larger precision means as low false positives, when the lower precision means high false positives. It is computed using the equation:

$$Precision = \frac{Tp}{Tp + FN} \quad (2)$$

Recall helps us to measure how many cases are predicted correctly as belongs to given category. The recall is also known as true positive rate (TP) is a proportion of positive cases which are correctly identified. It is computed using the equation:

$$Recall = \frac{Tp}{Tp + FN} \quad (3)$$

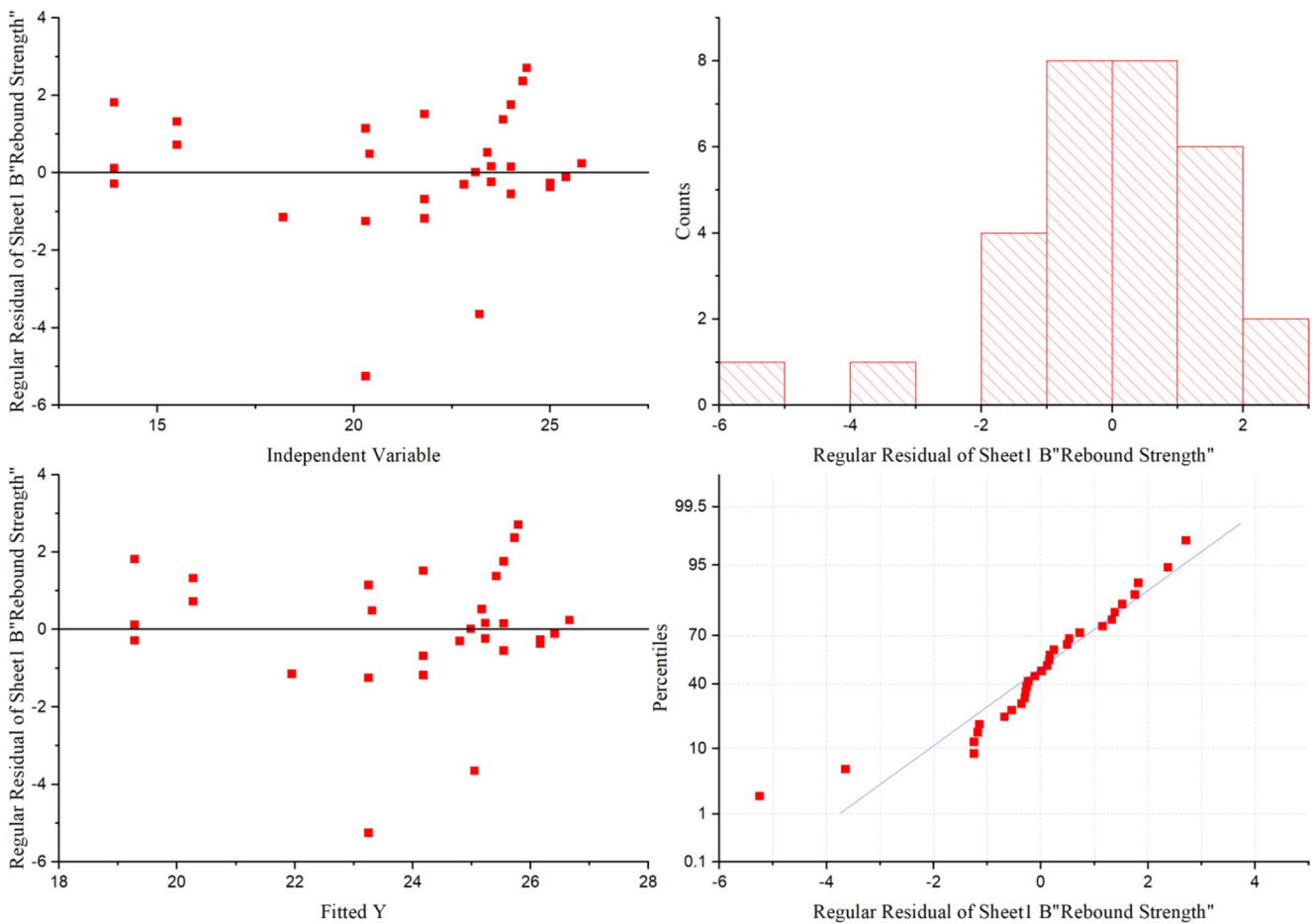


Fig. 9 Variation of rebound strength of ambient cured geopolymer concrete

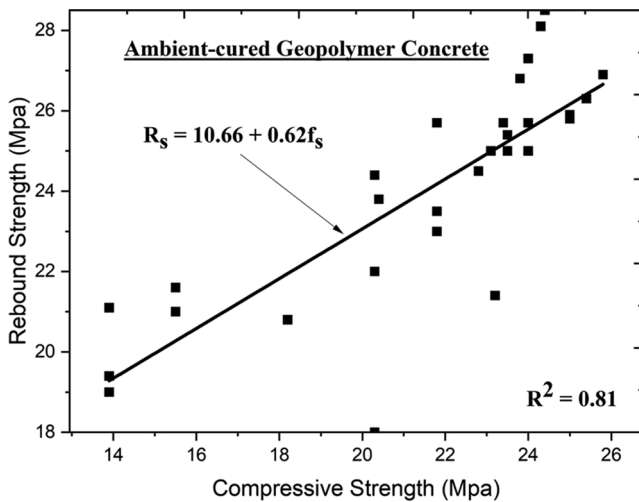


Fig. 10 Linear regression between compressive strength and rebound strength of ambient-cured GPC

i.e., Tp: True Positive, Tn: True Negative Fn: False Negative, Fp: False Positive.

F-1 score also called as F-Score or F-Measure. F1 score is the measure of calculating the weighted average of precision and recall. It is the weighted harmonic mean of the evaluation metric as precision and recall. F1 score ranges from 0 to 1 and F1 score is considered perfect when it is 1 which means that the model as low false positives and low false negatives. F1 score is calculated using the equation:

$$F-1 \text{ Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}).$$

AUC infers district under the curve so to discuss ROC AUC score we really have to be **depicted** ROC bend to begin **with**. It is a diagram that envisions the **trade-off** between clear beyond any doubt rate (TPR) and fake positive rate (FPR). In a common sense, for each **cut-off**, we handle TPR and FPR and plot it on one chart. Clearly, the higher TPR and the lower FPR is for each edge the better consequently classifiers that have twists that are more upper left-side are **better**.

PR AUC Correspondingly to receiver operating characteristic curve and Area under the ROC Curve to characterize PR AUC it is to characterize in the Precision-Recall bend. It

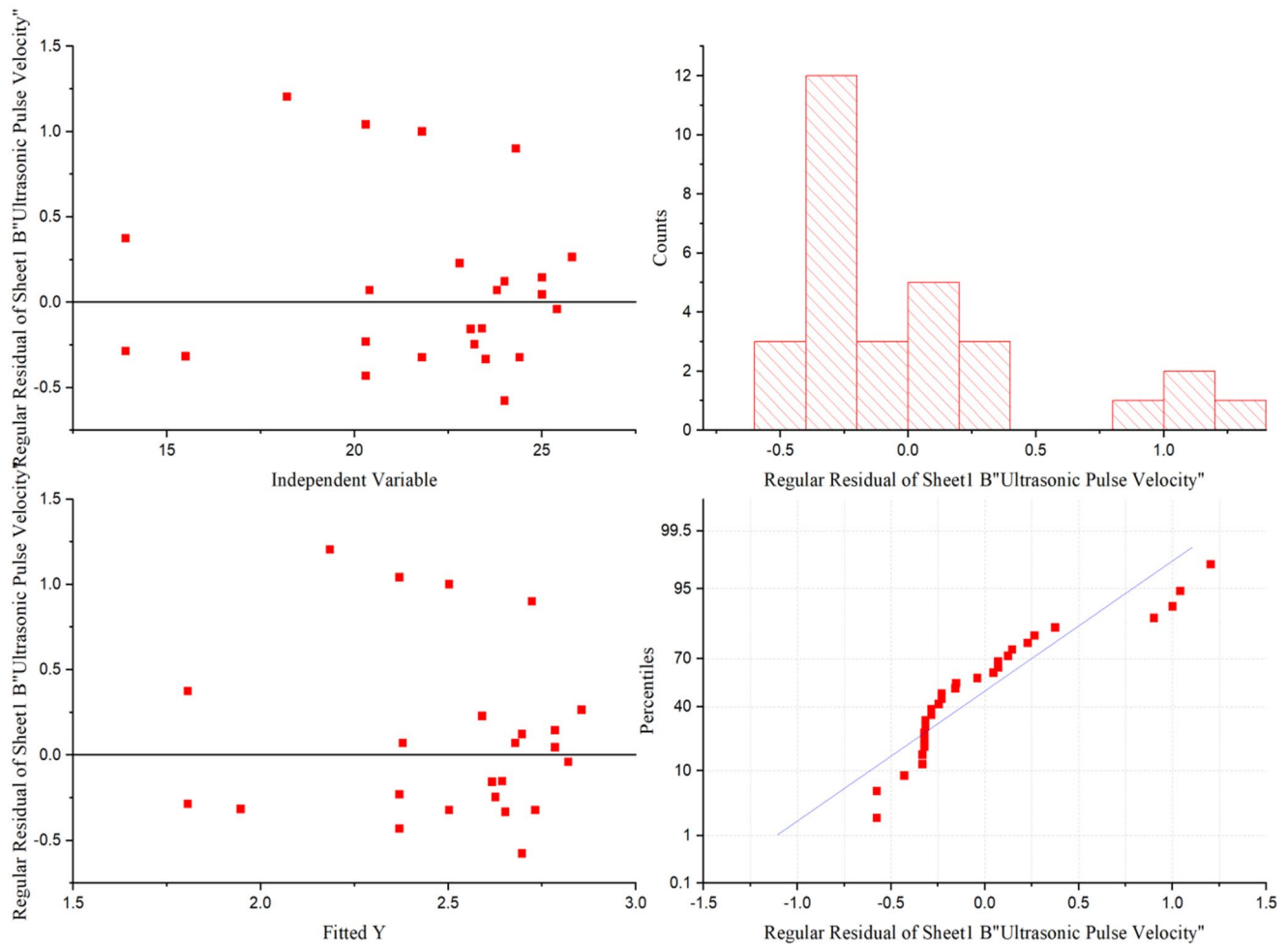


Fig. 11 Variation of ultrasonic pulse velocity of ambient cured geopolymer concrete

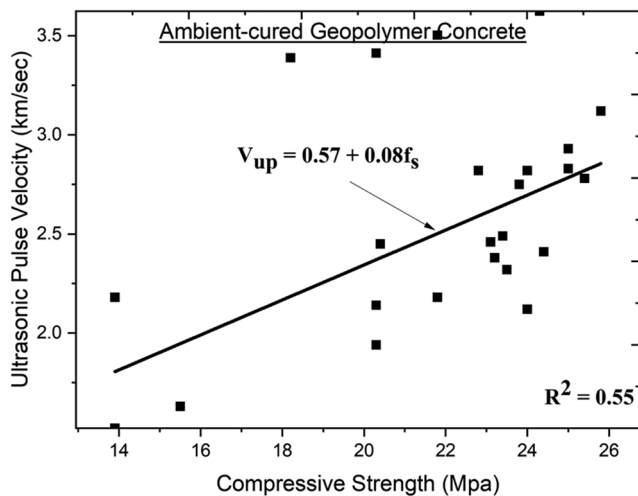


Fig. 12 Linear regression between compressive strength and rebound strength of ambient-cured GPC

is a bend that joins accuracy PPV & TPR in a solitary perception. For each limit, ascertain TPR& PPV value can get. **For the display** model better y center should be high. ROC AUC plot can utilize to create a taught choice with regard to the amazing precision/audit circumstance (Kaveh & Iranmanesh, 1998; Kaveh & Khalegi, 1998; Kaveh et al., 2008; Kaveh, 2017; Kaveh & Dadras, 2018). Clearly, the higher the review the lower the precision. for the negative lesson, it **maintains** high recall & precision review nearly all through the whole extent of edges. Figure 13 describes the compressive strength of brick, conventional concrete and geo polymer concrete cubes via non-destructive tests Data extraction and management. Figure 14 depicts the architecture of the ANN (RBF classifier) model used in this study. Figure 15 describes the methodology of development of an efficient artificial neural network (ANN) model (Kaveh & Servati, 2001; Kaveh et al., 2014, 2020; Kaveh & Maniat, 2015; Kaveh & Khavaninzadeh, 2023). Figure 16 shows the graphical representation of Bricks (In blue colour), Oven-cured GPC (in red colour) and ambient-cured GPC

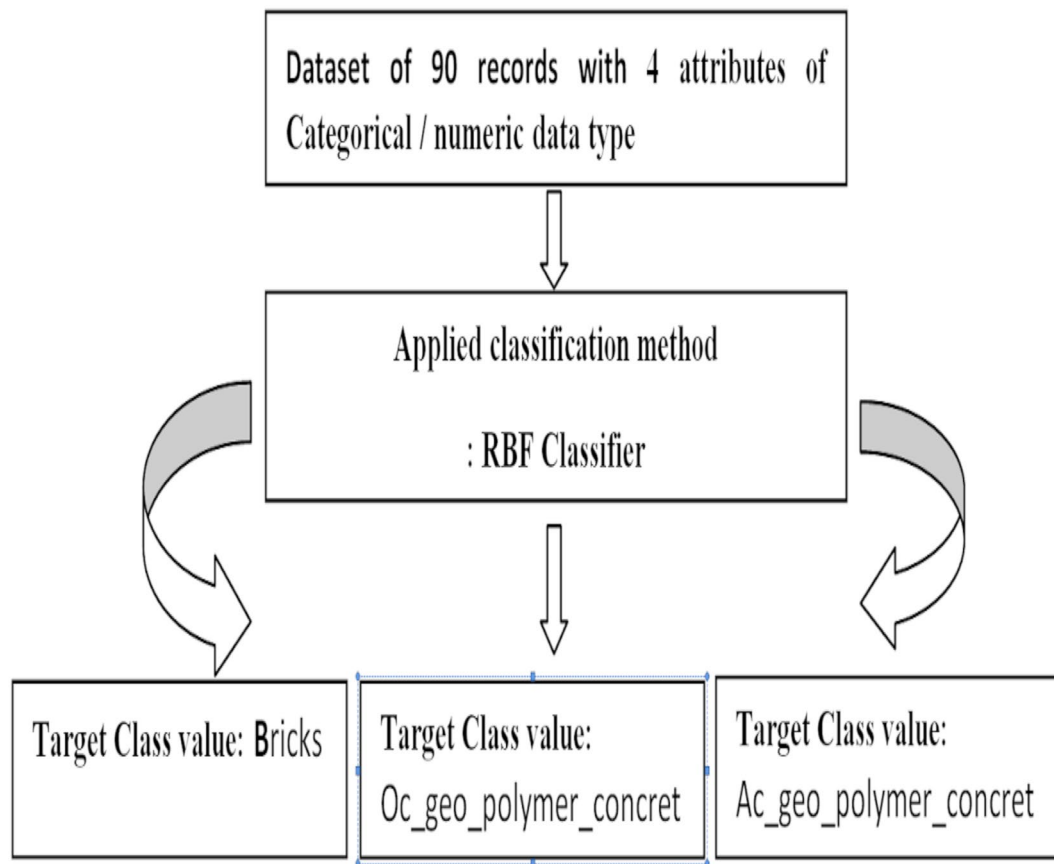


Fig. 13 Flow chart of working model

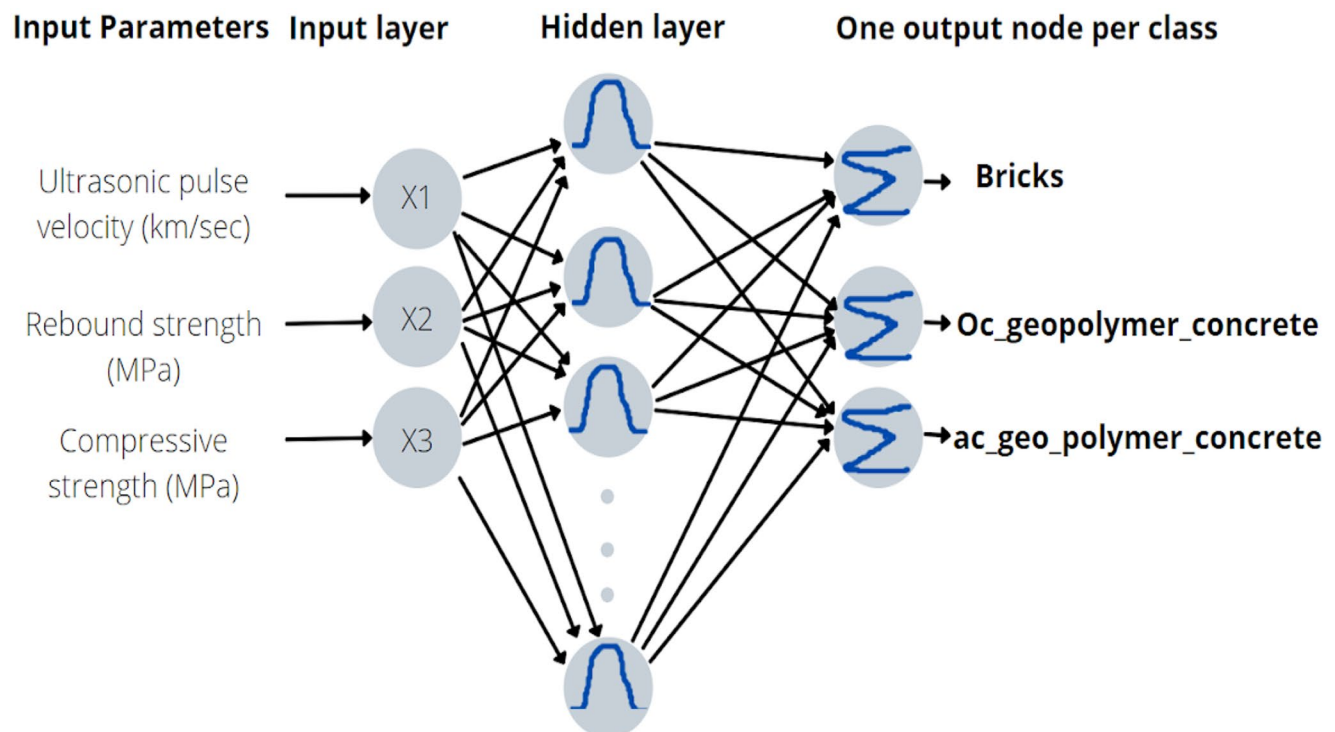


Fig. 14 Architecture of the ANN (RBF classifier) model used in this study

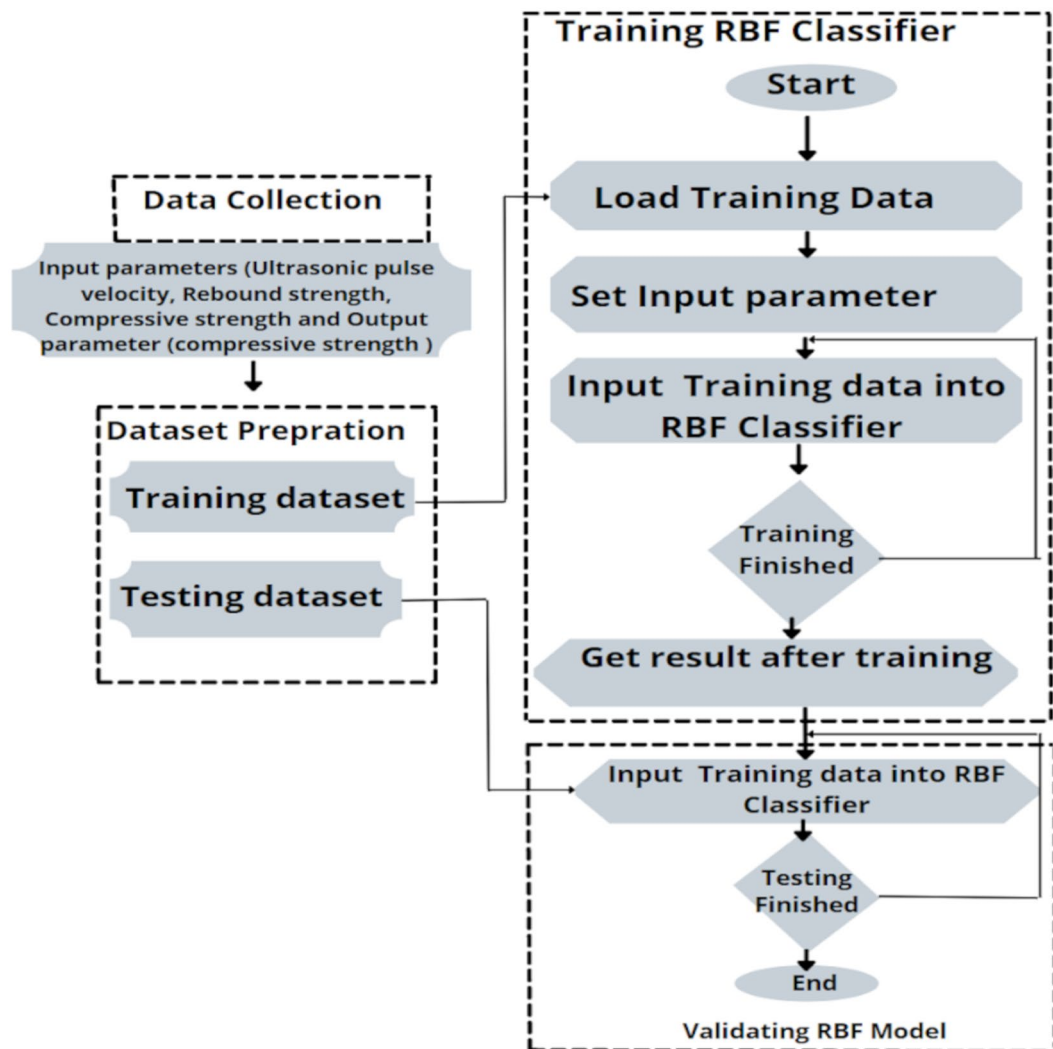


Fig. 15 Methodology of development of an efficient artificial neural network (ANN) model

(In Ocean blue colour). Table 4 describes the details of the results found after testing. Table 5 elaborates on the confusion matrix where we take predicted class and actual class values. Sum of the value is represented with n .

The Radial Basis Function (RBF) model is a powerful machine learning technique commonly used for function approximation, regression, and classification tasks. It is a type of artificial neural network (ANN) that employs radial basis functions as its activation functions. The RBF model is particularly known for its ability to model complex, non-linear relationships in data, making it effective for prediction tasks. The RBF model uses radial basis functions as the basis for transforming input data. A radial basis function is a real-valued function whose value depends only on the distance from a central point (the center). The most commonly used RBF is the Gaussian function:

$$\varphi(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

where r is the distance between the input and the center, and σ is a parameter that controls the width of the Gaussian.

The RBF network consists of three layers: (1) Takes in the input data, with each feature corresponding to a node. (2) Contains neurons, each associated with an RBF. Each neuron computes the distance between the input and its center and then applies the RBF to produce an output. (3) Produces the final prediction by combining the outputs of the hidden layer, typically using a weighted sum (Upreti et al., 2022; Sharma et al., 2023; Jayanthi et al., 2024; Nigam & Verma, 2024; Sharma et al., 2024).

Training an RBF network involves two main tasks: (1) The centers of the RBFs are chosen from the training data, either randomly or by clustering techniques like **K-Means**. (2) After the centers are fixed, the weights connecting the

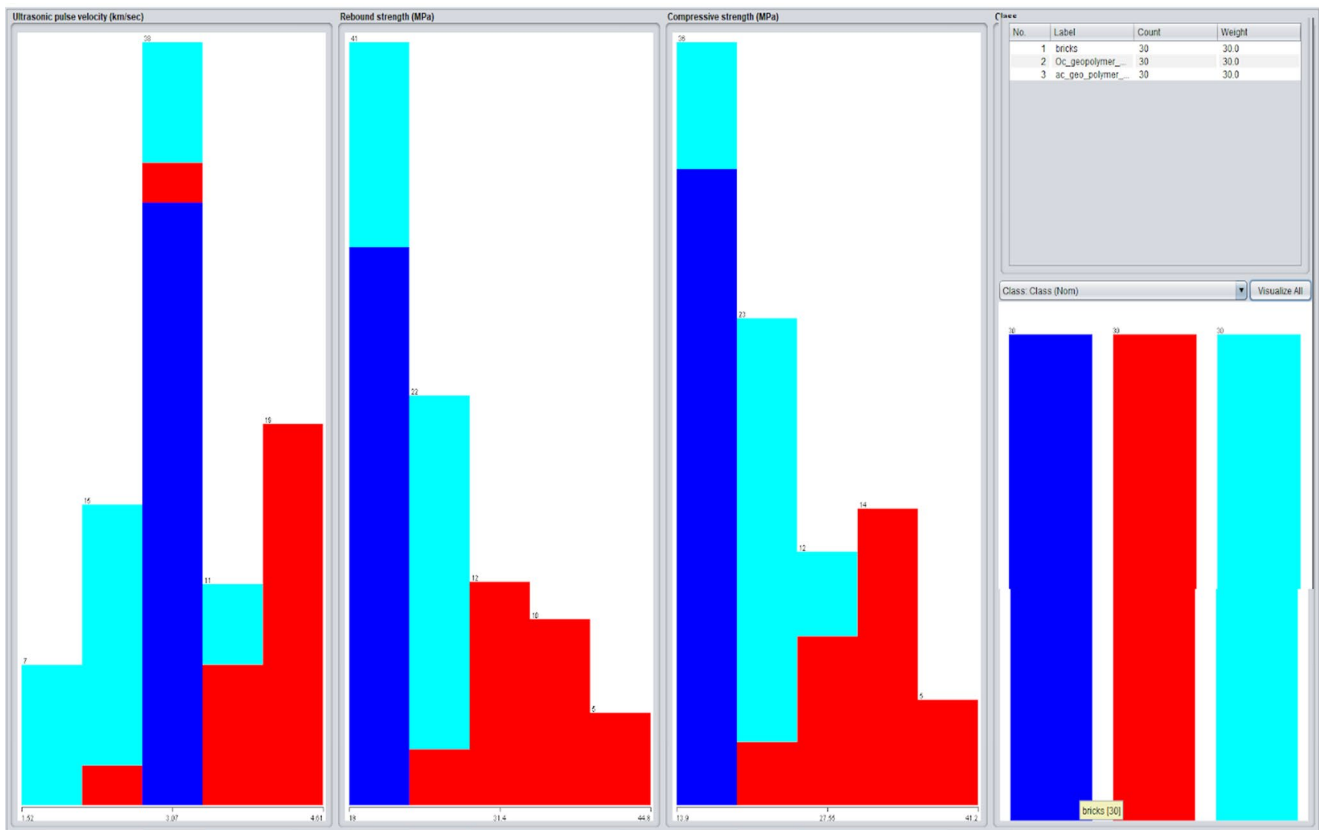


Fig. 16 Bricks (In blue colour), Oven-cured GPC (in red colour) and ambient-cured GPC (In Ocean blue colour)

Table 4 Results details

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
1.000	0.017	0.968	1.000	0.984	0.976	1.000	1.000	bricks
0.867	0.017	0.963	0.867	0.912	0.874	0.982	0.973	Oven-cured GPC
0.933	0.067	0.875	0.933	0.903	0.853	0.980	0.955	ambient-cured GPC
0.933	0.033	0.935	0.933	0.933	0.901	0.987	0.976	Weighted Avg.

Table 5 Confusion matrix where we take predicted class and actual class values. Sum of value is represented with n

Algorithm	Confusion Matrix				
RBF Classifier	$N=90$	Pre-dicted: bricks	Pre-dicted: Oven-cured GPC	Predicted: Ambient-cured GPC	
	Actual: bricks	30	0	0	30
	Actual: Oven-cured GPC	0	26	4	30
	Actual: Ambient-cured GPC	1	1	28	30
		31	27	32	

hidden layer to the output layer are learned using optimization techniques like **least squares** or **gradient descent**.

In function approximation or regression tasks, an RBF model can be used to predict continuous values based on input data. The model captures the underlying structure of

the data by adjusting the RBF centers and weights to minimize prediction errors. RBF networks can also be applied to classification problems, where the goal is to assign input data to predefined categories. The outputs are usually mapped to class probabilities, and the class with the highest probability is selected as the prediction.

The RBF network is particularly useful for problems where the relationship between input and output is non-linear. The non-linearity of RBFs allows the network to approximate complex functions effectively. RBFs produce outputs that are sensitive to nearby data points (due to the radial basis function's dependence on distance), making them suitable for modelling data that changes locally. Since the RBF network can separate the process of determining the centres (which can be done using clustering techniques) from learning the weights, it often results in faster training compared to traditional neural networks (Kumar et al., 2022,

2023; Kumar et al., 2022; Upreti & Verma, 2022; Nigam & Verma, 2023).

The performance of an RBF model heavily depends on the number and position of the centres. Too many centres can lead to overfitting, while too few centres might result in underfitting. For large datasets, the computation of distances between each input point and every center becomes expensive, making the RBF network less scalable compared to some other models like support vector machines (SVM) or deep neural networks.

Time Series Prediction: RBF networks are widely used in time series forecasting due to their ability to model non-linear temporal patterns. **Function Approximation:** They are also used in approximating complex functions in fields like finance, physics, and engineering. **Classification Tasks:** RBF models are employed in classification tasks such as pattern recognition, image classification, and medical diagnosis.

The RBF model is a versatile and powerful tool in machine learning, especially for predicting non-linear relationships in data. While it offers advantages like localized response and fast training, careful consideration is needed regarding the number of centers and their positions to avoid overfitting or underfitting. With appropriate tuning, the RBF model can be highly effective in tasks like regression, classification, and time series prediction. Table 5 represent a confusion matrix where we take predicted class and actual class values. Sum of the value is represented with n. **Correctly Classified Instances are 93.3333% (84/90) whereas incorrectly Classified Instances are 6.6667%.**

Conclusion

This study predicts compressive strength by using useful features of compressive strength. The prediction algorithm plays a crucial role in the decision-making process to use the right material.

- The proposed framework helps to save time and money by predicting their Compressive strength based on parameters. This study investigates the reliability estimation of the compressive strength of brick, conventional concrete, and geopolymers concrete cubes through the application of non-destructive tests (NDTs). Non-destructive testing methods, such as rebound hammer and ultrasonic pulse velocity (UPV) tests, offer a rapid and practical approach for assessing the structural integrity and strength of materials without causing damage.
- The research evaluates the effectiveness of these methods in estimating the compressive strength of the three different materials—brick, conventional concrete, and

geopolymer concrete—and compares the results with conventional destructive testing (CDT) methods like the compressive strength test on cube specimens. Statistical analysis is employed to establish correlations between NDT results and actual compressive strength values, providing insight into the reliability and accuracy of NDT techniques for different materials.

- The findings demonstrate that NDTs can serve as reliable alternatives to CDTs, though the level of accuracy varies depending on the material type. The study concludes that while NDT methods are highly effective for preliminary evaluations and in-situ assessments, further calibration may be required for different materials to ensure precise compressive strength estimation.

Author contributions MV drafted and wrote the paper work in the manuscript.

Funding The authors have not disclosed any funding.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

References

- Abd, A. (2014). 'Reliability of using nondestructive tests to estimate compressive strength of building stones and bricks'. *Alexandria Engineering Journal*, 51(3), 193–203. Available at: <https://doi.org/10.1016/j.aej.2012.05.004>
- ACI 228.1R-03 (2003). 'In-Place Methods to Estimate Concrete Strength Reported', ACI Committee Reports, (228, 1R), p. 44.
- ASTM C597 16 (2016). 'Standard Test Method for Pulse Velocity Through Concrete ASTM C 597', ASTM International, p. 4. Available at: <https://doi.org/10.1520/C0597-16.2>
- ASTM C805/C805M 18 (2002). 'Rebound Number of Hardened Concrete', ASTM International, (2019), pp. 2–4. Available at: <https://doi.org/10.1520/C0805>
- Breysse, D. (2012). 'Nondestructive evaluation of concrete strength: An historical review and a new perspective by combining NDT methods'. *Construction and Building Materials*, 33, 139–163. Available at: <https://doi.org/10.1016/j.conbuildmat.2011.12.103>
- de Rodriguez, G., Rodriguez Viacava, I., & Aguado, A. (2016). 'Simple and Rational Methodology for the Formulation of Self-Compacting Concrete Mixes'. *Journal of Materials in Civil Engineering*, 28(2), 04015116.
- Erdem, S., Hanbay, S., & Güler, Z. (2018). 'Micromechanical damage analysis and engineering performance of concrete with colloidal nano-silica and demolished concrete aggregates'. *Construction and Building Materials*, 171, 634–642. Available at: <https://doi.org/10.1016/j.conbuildmat.2018.03.197>
- Guo, X., Shi, H., & Dick, W. A. (2010). 'Cement & Concrete Composites Compressive strength and microstructural characteristics of class C fly ash geopolymer'. *Cement and Concrete Composites*, 32(2), 142–147. Available at: <https://doi.org/10.1016/j.cemconcomp.2009.11.003>

- Jayanthi, N. (2024). 'Length and width of low-light, concrete hairline crack detection.pdf'. *Asian Journal of Civil Engineering*, 25(3), 2705–2714. Available at: <https://doi.org/10.1007/s42107-023-00939-0>
- Kaveh, A. (2017). Applications of metaheuristic optimization algorithms in civil engineering, Applications of Metaheuristic Optimization Algorithms in Civil Engineering. Available at: <https://doi.org/10.1007/978-3-319-48012-1>
- Kaveh, A., & Dadras, A. (2018). 'Structural damage identification using an enhanced thermal exchange optimization algorithm'. *Engineering Optimization*, 50(3), 430–451. Available at: <https://doi.org/10.1080/0305215X.2017.1318872>
- Kaveh, A., & Iranmanesh, A. (1998). Comparative study of backpropagation and improved counterpropagation neural Nets in structural analysis and optimization. *International Journal of Space Structure*, 13(4), 177–185.
- Kaveh, A., & Khalegi, A. (1998). 'Prediction of strength for concrete specimens using artificial neural networks', *Asian Journal of Civil Engineering*, 2, pp. 1–13. Available at: https://hero.epa.gov/hero/index.cfm/reference/details/reference_id/6859258 (Accessed: 17 April 2023).
- Kaveh, A., & Khavaninzadeh, N. (2023). Efficient training of two ANNs using four meta-heuristic algorithms for predicting the FRP strength. *Structures*, 52(March), 256–272.
- Kaveh, A., & Maniat, M. (2015). 'Damage detection based on MCSS and PSO using modal data'. *Smart Structures and Systems*, 15(5), 1253–1270. Available at: <https://doi.org/10.12989/sss.2015.15.5.1253>
- Kaveh, A., & Servati, H. (2001). *Design of double layer grids using backpropagation neural networks* (Vol. 79, pp. 1561–1568). Computers & Structures.
- Kaveh, A., Gholipour, Y., & Rahami, H. (2008). Optimal design of transmission towers using genetic algorithm and neural networks. *International Journal of Space Structure*, 23(1), 1–20.
- Kaveh, A., Javadi, S. M., & Maniat, M. (2014). Damage assessment via modal data with a mixed particle swarm strategy, ray optimizer, and harmony search. *Asian Journal of Civil Engineering*, 15(1), 95–106.
- Kaveh, A., Izadifard, R. A., & Mottaghi, L. (2020). 'Optimal design of planar RC frames considering CO2 emissions using ECBO, EVPS and PSO metaheuristic algorithms'. *Journal of Building Engineering*, 28, 101014. Available at: <https://doi.org/10.1016/J.JOBE.2019.101014>
- Kheder, G. F. (1999). 'Two stage procedure for assessment of in situ concrete strength using combined non-destructive testing'. *Materials and Structures/Materiaux et Constructions*, 32(6), 410–417. Available at: <https://doi.org/10.1007/bf02482712>
- Kim, J. K. (2009). 'Effect of carbonation on the rebound number and compressive strength of concrete'. *Cement and Concrete Composites*, 31(2), 139–144. Available at: <https://doi.org/10.1016/j.cemconcomp.2008.10.001>
- Kumar, R. (2022). 'Influence of chloride and sulfate solution on the long-term durability of modified rubberized concrete'. *Journal Of Applied Polymer Science*, 2022(139), 1–15. Available at: <https://doi.org/10.1002/app.52880>
- Kumar, N. (2023). 'Environmental Concern in TPB Model for Sustainable IT Adoption'. In *International Conference on Information Systems and Intelligent Applications*, Lecture Notes in Networks and Systems 550, pp. 59–70. Available at: https://doi.org/10.1007/978-3-031-16865-9_5
- Kumar, R., Verma, M., & Dev, N. (2022). 'Investigation on the Effect of Seawater Condition, Sulphate Attack, Acid Attack, Freeze-Thaw Condition, and Wetting-Drying on the Geopolymer Concrete'. *Iranian Journal of Science and Technology Transactions of Civil Engineering*, 46(4), 2823–2853. Available at: <https://doi.org/10.1007/s40996-021-00767-9>
- Kumar, R., Verma, M., & Dev, N. (2024). 'Analysis of PCE-based Superplasticiser for the Different Types of Cement using Marsh Cone Test'. *Evergreen*, 11(2), 665–672. Available at: <https://doi.org/10.5109/7183337>
- Mohan, A., & Poobal, S. (2018). 'Crack detection using image processing: A critical review and analysis'. *Alexandria Engineering Journal*, 57(2), 787–798. Available at: <https://doi.org/10.1016/j.aej.2017.01.020>
- Mukharjee, B. B., & Barai, S. V. (2020). 'Influence of Incorporation of Colloidal Nano-Silica on Behaviour of Concrete'. *Iranian Journal of Science and Technology - Transactions of Civil Engineering*, 44(2), 657–668. Available at: <https://doi.org/10.1007/s40996-020-00382-0>
- Nigam, M., & Verma, M. (2023). 'Effect of Nano-Silica on the Fresh and Mechanical Properties of Conventional Concrete'. *Forces in Mechanics*, 10(22), 100165. Available at: <https://doi.org/10.1016/j.finmec.2022.100165>
- Nigam, M., & Verma, M. (2024). 'Prediction of compressive strength of nano-silica concrete by using random forest algorithm', *Asian Journal of Civil Engineering [Preprint]*, (0123456789). Available at: <https://doi.org/10.1007/s42107-024-01107-8>
- Nigam, M., & Verma, M. (2025). Mechanical strength prediction of Nano-Silica concrete composites using machine learning techniques. *Journal of Polymer & Composites*, 13(1), 963–973.
- O, K., Z, R. and J-C, R. (2000). Evaluation of mechanical rock properties using a Schmidt hammer. *International Journal of Rock Mechanics and Mining Sciences*, 37, 723–728.
- Qaswari, H. (2000). Concrete strength by combined nondestructive methods simply and reliably predicted. *Cement and Concrete Research*, 30(2000), 739–746.
- Sharma, U. (2024). 'Behavior of Fibers in Geopolymer Concrete: A Comprehensive Review'. *Buildings*, 14(136), 1–28. Available at: <https://doi.org/10.3390/buildings14010136>
- Sharma, U., Gupta, N., & Verma, M. (2023). 'Prediction of compressive strength of GGBFS and Flyash-based geopolymer composite by linear regression, lasso regression, and ridge regression'. *Asian Journal of Civil Engineering*, 24(8), 3399–3411. Available at: <https://doi.org/10.1007/s42107-023-00721-2>
- Solis-Carcano, R., & Moreno, E. I. (2008). 'Evaluation of concrete made with crushed limestone aggregate based on ultrasonic pulse velocity'. *Construction and Building Materials*, 22(6), 1225–1231. Available at: <https://doi.org/10.1016/j.conbuildmat.2007.0.1014>
- Sreenivasulu, C., Guru Jawahar, J., & Sashidhar, C. (2020). 'Effect of Copper Slag on Micro, Macro, and Flexural Characteristics of Geopolymer Concrete'. *Journal of Materials in Civil Engineering*, 32(5), 04020086.
- Szilágyi, K., Borosnyói, A., & Zsigovics, I. (2011). 'Rebound surface hardness of concrete: Introduction of an empirical constitutive model'. *Construction and Building Materials*, 25(5), 2480–2487. Available at: <https://doi.org/10.1016/j.conbuildmat.2010.11.070>
- Tennakoon, C. (2017). 'Chloride ingress and steel corrosion in geopolymer concrete based on long term tests'. *Materials and Design*, 116, 287–299. Available at: <https://doi.org/10.1016/j.matdes.2016.12.030>
- Tiwari, P. K., & Verma, M. (2024). Forecasting of crushing strength of sustainable concrete by employing deep and random forest machine learning. *Journal of Polymer & Composites*, 12(7), 85–90.
- Turgut, P., Yesilnacar, M. I., & Bulut, H. (2008). 'Physico-thermal and mechanical properties of Sanliurfa limestone, Turkey'. *Bulletin of Engineering Geology and the Environment*, 67(4), 485–490. Available at: <https://doi.org/10.1007/s10064-008-0145-2>
- Upreti, K. (2022). 'Prediction of Mechanical Strength by Using an Artificial Neural Network and Random Forest Algorithm'.

- Journal of Nanomaterials*, 2022, 1–12. Available at: <https://doi.org/10.1155/2022/7791582>
- Upreti, K., & Verma, M. (2022). 'Prediction of compressive strength of high-volume fly ash concrete using artificial neural network'. *Journal of Engineering Research and Application*, 1(2), 24–32. Available at: <https://doi.org/10.55953/JERA.2022.2104>
- Varghese, L., Kanta Rao, V. V. L., & Parameswaran, L. (2019). 'Nanosilica-added concrete: Strength and its correlation with time-dependent properties'. *Proceedings of Institution of Civil Engineers: Construction Materials*, 172(2), 85–94. Available at: <https://doi.org/10.1680/jcoma.17.00031>
- Vasconcelos, G. (2008). 'Ultrasonic evaluation of the physical and mechanical properties of granites'. *Ultrasonics*, 48(5), 453–466. Available at: <https://doi.org/10.1016/j.ultras.2008.03.008>
- Verma, M. (2021). *Study on behaviour of geopolymer concrete*. Delhi Technological University.
- Verma, M., & Dev, N. (2022). 'Effect of ground granulated blast furnace slag and fly ash ratio and the curing conditions on the mechanical properties of geopolymer concrete'. *Structural Concrete*, 23(4), 2015–2029. Available at: <https://doi.org/10.1002/suco.202000536>
- Xu, C., et al. (2014). An infrared thermal image processing framework based on superpixel algorithm to detect cracks on metal surface. *Infrared Physics and Technology*, 67, 266–272. <https://doi.org/10.1016/j.infrared.2014.08.002>. Available at:.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.